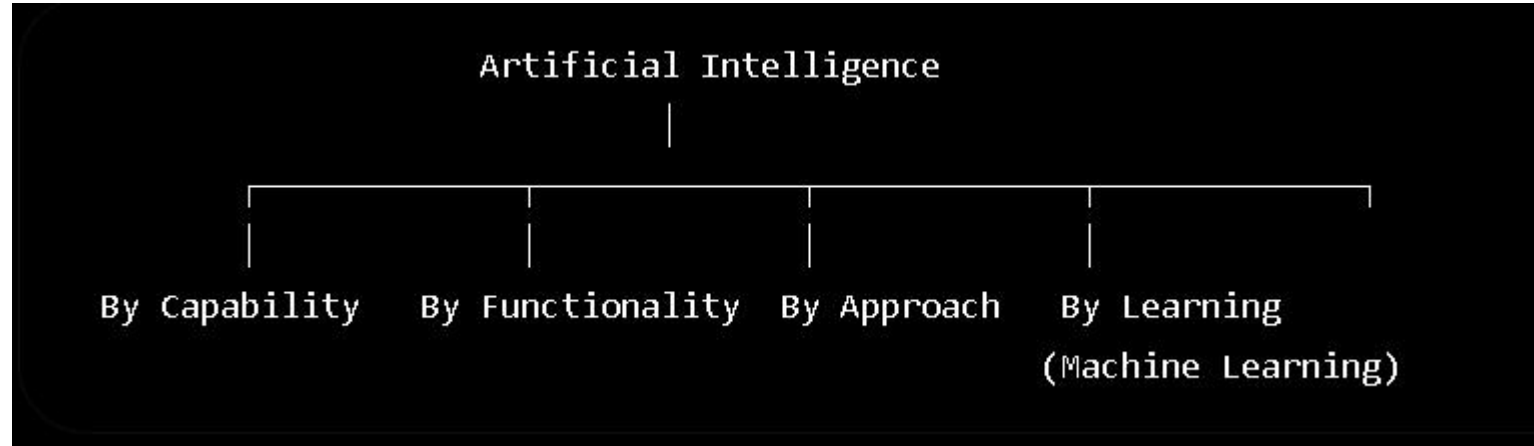


TYPES OF ARTIFICIAL INTELLIGENCE

Artificial Intelligence can be classified in four major ways:



AI has no single classification. Researchers classify AI from different perspectives. Today, we will study the four most widely accepted classifications used in academia and industry.

AI Classification by Capability

Capability-based classification categorizes AI according to its level of intelligence and problem-solving ability.

Three Types:

Type	Meaning	Current Status
1. ANI	Artificial Narrow Intelligence	✓ Exists
2. AGI	Artificial General Intelligence	✗ Under Research
3. ASI	Artificial Super Intelligence	✗ Theoretical

Artificial Narrow Intelligence (ANI)

Definition:

AI designed to perform one specific task exceptionally well.

Characteristics:

1. Performs a single task
2. Cannot think like humans
3. Cannot transfer knowledge
4. Domain-specific intelligence

Examples:

1. ChatGPT
2. Google Translate
3. Face Recognition
4. Spam Detection
5. Recommendation Systems

Artificial General Intelligence (AGI)

Definition:

AI capable of performing any intellectual task that a human can perform(Under Research).

Characteristics:

1. Learns new skills
2. Solves unfamiliar problems
3. Human-level reasoning
4. Transfers knowledge across domains

Example:

One AI can...

1. Code
2. Drive
3. Cook
4. Teach
5. Diagnose Diseases
6. Learn New Skills

Artificial Super Intelligence (ASI)

Definition:

AI that exceeds human intelligence in all domains.

Capabilities:

1. Scientific discovery
2. Medical breakthroughs
3. Self-improvement
4. Creative innovation
5. Strategic decision making

Current Status:

Hypothetical

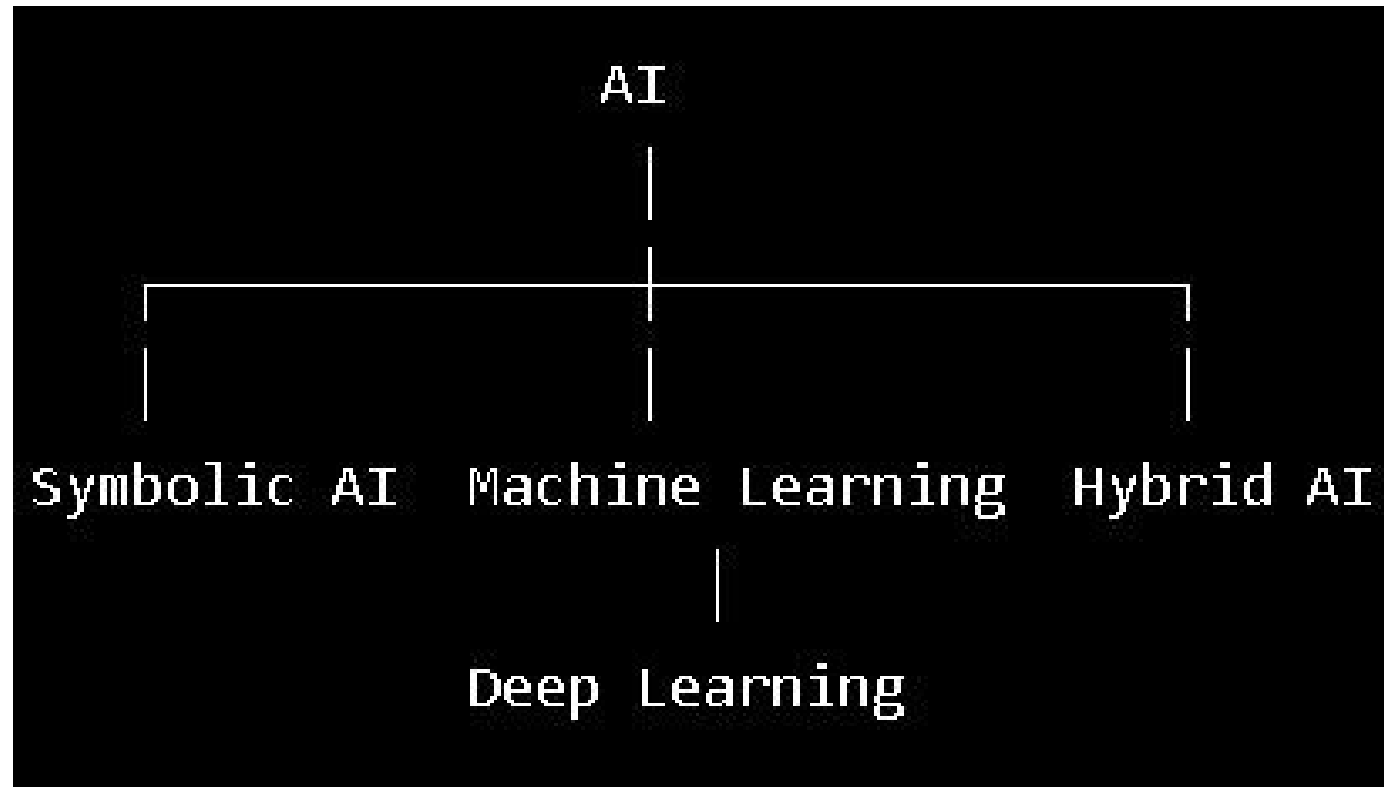
AI Classification by Functionality

This classification is based on how AI behaves and processes information.

Type	Description	Example
1. <i>Reactive Machines</i>	<i>No memory</i>	<i>IBM Deep Blue</i>
2. <i>Limited Memory</i>	<i>Uses past data</i>	<i>Self-driving Cars</i>
3. <i>Theory of Mind</i>	<i>Understands emotions</i>	<i>Under Research</i>
4. <i>Self-Aware AI</i>	<i>Conscious AI</i>	<i>Does Not Exist</i>

AI Classification by Approach

Classifies AI based on how the system is designed and implemented.



Symbolic AI

Also Known As:

1. Rule-Based AI
2. Knowledge-Based AI
3. Expert Systems

Example Rule:

IF Fever = Yes
AND Cough = Yes
THEN Flu = Possible

Works Using:

1. Facts
2. Rules
3. Logic
4. Knowledge Base
5. Inference Engine

Applications:

1. Medical Diagnosis
2. Banking Rules
3. Tax Systems
4. Legal Systems

Hybrid AI

- Hybrid AI is an AI approach that combines two or more AI techniques to build systems that are more accurate, reliable, and intelligent than using a single technique alone.
 - Most modern AI systems are hybrid because no single AI approach is best for every problem.
 - **Hybrid AI = Rule-Based AI + Learning-Based AI**
- or
- **Hybrid AI combines symbolic reasoning with machine learning or deep learning to leverage the strengths of both approaches.**

Ex.: A self-driving car uses multiple AI techniques together.

Deep Learning:

- Detect pedestrians
- Detect traffic signs
- Detect lanes

Symbolic AI:

Traffic rules such as:

IF Traffic Light = Red
THEN Stop
IF Speed > Limit
THEN Slow Down

Together, they create a safer driving system.

AI Classification by Learning Method

	Learning Type	Data Required	Example
1.	Supervised	Labeled	House Price Prediction
2.	Unsupervised	Unlabeled	Customer Segmentation
3.	Semi-Supervised	Small Labeled + Large Unlabeled	Medical Images
4.	Self-Supervised	Data Creates Labels	GPT, BERT
5.	Reinforcement	Reward & Penalty	Robotics, Games